Assignment 2: EC5644

Ronan Engel

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1 Question 1

This question utilizes the following gaussian distributions to define the dataset and sample space. They describe two classes, L = 0 and L = 1, both of which are made of equally weighted gaussian distributions. They are as follows.

$$\mathbf{m}_{01} = \begin{bmatrix} -0.9 \\ -1.1 \end{bmatrix}, \quad \mathbf{m}_{02} = \begin{bmatrix} 0.8 \\ 0.75 \end{bmatrix}, \quad \mathbf{m}_{11} = \begin{bmatrix} -1.1 \\ 0.9 \end{bmatrix}, \quad \mathbf{m}_{12} = \begin{bmatrix} 0.9 \\ -0.75 \end{bmatrix}$$

$$\mathbf{C}_{ij} = \begin{bmatrix} 0.75 & 0 \\ 0 & 1.25 \end{bmatrix} \quad \text{for all } \{i, j\} \text{ pairs}$$

1.1 Theoretically Optimal Classifier

Determining the optimal classifier based upon the knowledge of a true PDF is done through the following derivation of the following decision rule, following from the derivation used in Assignment 1, Section 1.1:

$$\begin{aligned} D(x) &= \arg\min \, R(D = d \mid x) \\ &= \arg\min \left(\lambda_{d0} \, P(L = 0 \mid x) + \lambda_{d1} \, P(L = 1 \mid x) \right) \end{aligned}$$

Decide $D(x) = 1$ when $Risk(D = 1 \mid x) < Risk(D = 0 \mid x)$:

$$\frac{P(L = 1 \mid x)}{P(L = 0 \mid x)} > \frac{\lambda_{10} - \lambda_{00}}{\lambda_{01} - \lambda_{11}}$$

Apply Bayes Rule:

$$\frac{P(x \mid L = 1)}{P(x \mid L = 0)} > \frac{(\lambda_{10} - \lambda_{00}) \, P(L = 0)}{(\lambda_{01} - \lambda_{11}) \, P(L = 1)}$$

Assume that making a wrong decision costs more than making a correct decision.
Given: $\lambda_{10} = 1, \ \lambda_{00} = 0, \ \lambda_{01} = 1, \ \lambda_{11} = 0, \ P(L = 0) = 0.6, \ P(L = 1) = 0.4$

$$\frac{P(x \mid L = 1)}{P(x \mid L = 0)} > \frac{(1) \, P(L = 0)}{(1) \, P(L = 1)}$$

$$\gamma = \frac{0.6}{0.4}$$

$$\gamma = 1.5$$

Using this optimal classifier, this decision rule of $\gamma = 1.5$ can be applied to all samples, deriving the following ROC curve. Furthermore, γ can also derived empirically, through the comparison of various decision thresholds. Described in Figure 1, the ROC curve of the optimal classifier can be seen, finding an error rate of **0.2789** at $\gamma = \mathbf{1.50}$ and an empirically derived error of **0.2768** at $\gamma = \mathbf{1.59}$.

Furthermore, utilizing the optimal classification rule, the following graph, Figure 2, displays the classification of the samples according to the theoretically optimal log decision boundary, validated on 10,000 samples distributed according to the governing Gaussians.

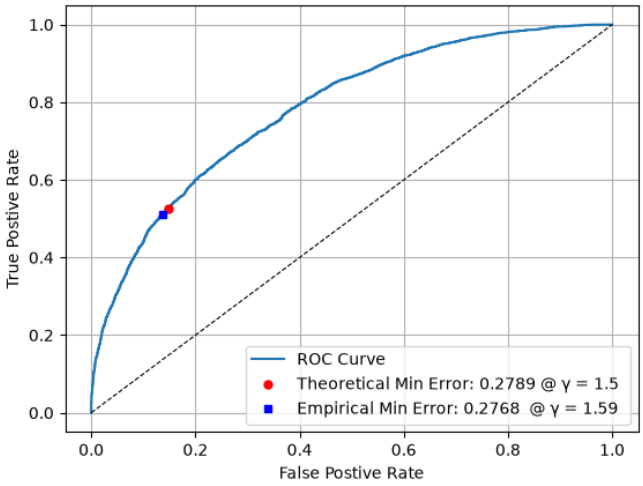


Figure 1: ERM Classifier ROC Curve.

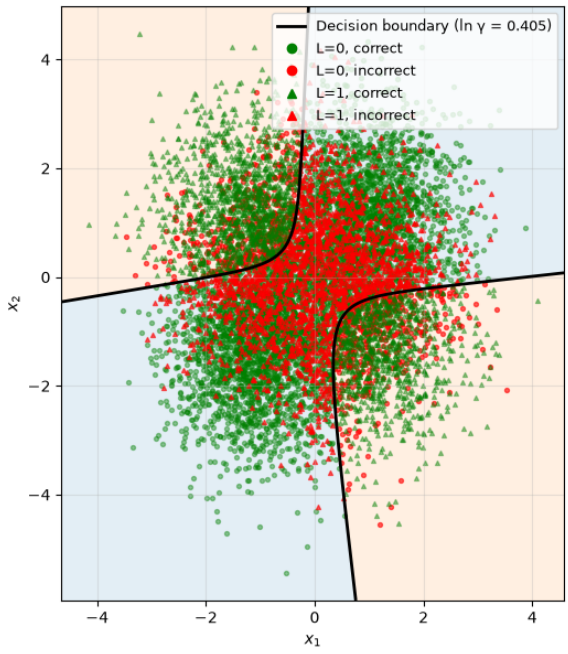


Figure 2: Ideal Classifications with 10,000 Samples.

1.2 Parameter Estimation

For this part, three logistic-function-based approximations of the class posterior are trained using Maximum Likelihood (ML) parameter estimation, one for each of the training sets $\mathcal{D}^{20}_{\text{train}}$, $\mathcal{D}^{200}_{\text{train}}$, and $\mathcal{D}^{2000}_{\text{train}}$, comprised of 20, 200, and 2000 samples generated from the Gaussian mixtures described in Section 1. Rather than modeling each class-conditional pdf, the class posterior is approximated directly by the logistic function in Equation 1, where $\mathbf{z}(\mathbf{x})$ is the linear or quadratic feature vector.

$$P(L=1 \mid \mathbf{x}) \approx h(\mathbf{x}, \mathbf{w}) = \frac{1}{1+e^{-\mathbf{w}^\top \mathbf{z}(\mathbf{x})}}, \qquad \mathbf{z}_{\text{lin}}(\mathbf{x}) = [1,\, x_1,\, x_2]^\top, \quad \mathbf{z}_{\text{quad}}(\mathbf{x}) = [1,\, x_1,\, x_2,\, x_1^2,\, x_1x_2,\, x_2^2]^\top \tag{1}$$

Since each training label $l_n \in \{0, 1\}$ is Bernoulli with probability $h(\mathbf{x}_n, \mathbf{w})$ and the samples are iid, the ML estimate of the weights minimizes the negative log-likelihood of the training labels, described in Equation 2.

$$\hat{\mathbf{w}}_{\text{ML}} = \arg\min_{\mathbf{w}} \; -\frac{1}{N}\sum_{n=1}^{N}\Big[\, l_n \ln h(\mathbf{x}_n, \mathbf{w}) + (1-l_n)\ln\big(1-h(\mathbf{x}_n, \mathbf{w})\big)\Big] \tag{2}$$

There is no closed-form solution, so Equation 2 is minimized through gradient descent, described in Equation 3. Starting from $\mathbf{w}^{(0)} = \mathbf{0}$, a fixed step size of $\eta = 4/\lambda_{\max}\big(\frac{1}{N}\mathbf{Z}^\top\mathbf{Z}\big)$ is used, which guarantees the NLL decreases each iteration, and the iterations stop once the norm of the gradient falls below 10^{-6}.

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - \eta\,\frac{1}{N}\,\mathbf{Z}^\top\Big(h\big(\mathbf{Z}\mathbf{w}^{(k)}\big) - \mathbf{l}\Big) \tag{3}$$

To approximate the minimum-P(error) classification rule, each validation sample is assigned the class with the larger approximate posterior, the MAP decision rule. Since $h(\mathbf{x}, \hat{\mathbf{w}}_{\text{ML}})$ approximates $P(L=1 \mid \mathbf{x})$ directly, this means deciding $D(\mathbf{x}) = 1$ when $h(\mathbf{x}, \hat{\mathbf{w}}_{\text{ML}}) > 0.5$, or equivalently when $\hat{\mathbf{w}}_{\text{ML}}^\top \mathbf{z}(\mathbf{x}) > 0$. Unlike Part 1, no threshold γ is needed, since the class priors are absorbed into the bias weight w_0 during training. Applying these trained classifiers to $\mathcal{D}^{10\text{K}}_{\text{validate}}$ yields the results in Table 1.

1.3 Results & Discussion

The relationship between the classification, the models employed, and results are revealing to both training sets and model selection considerations. Viewing Figure 3, the logistic-linear classifiers, unable to effectivity cut the feature space, leading to error being approximately the side of of the prior for Class 1. This is especially exemplified in Figure 3, which is self evident in the limitations of the linear discriminant in classifying the data.

Model	N	Iterations	Final NLL	$\hat{P}$(error)
Linear	20	28	0.5403	0.3999
Linear	200	22	0.6692	0.4033
Linear	2000	21	0.6679	0.4023
Quadratic	20	1031	0.3084	0.3368
Quadratic	200	476	0.5727	0.2941
Quadratic	2000	529	0.5418	0.2868

Table 1: Logistic-linear and logistic-quadratic classifiers trained on $\mathcal{D}_{\text{train}}^{N}$, with $\hat{P}$(error) estimated from error counts on $\mathcal{D}_{\text{validate}}^{\text{10K}}$.

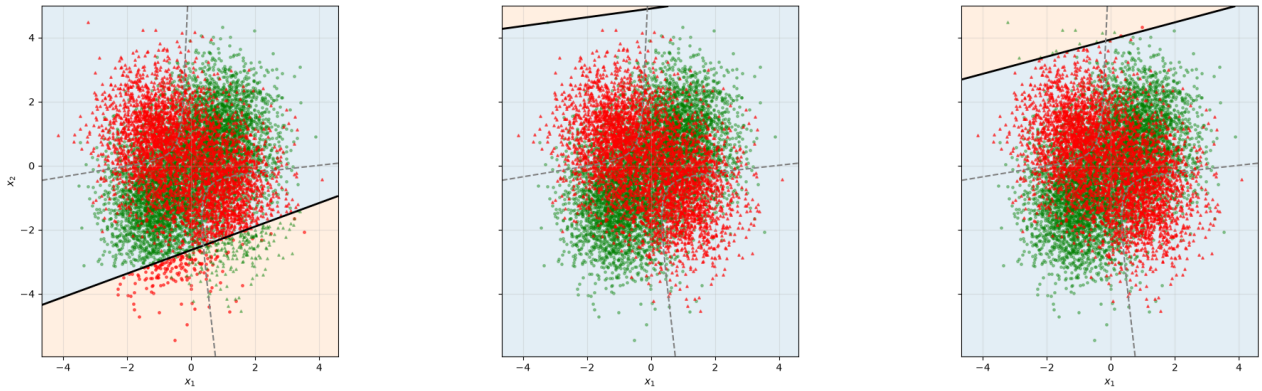


Figure 3: Linear Classification Boundaries of D_{train}^{20}, D_{train}^{200}, and D_{train}^{2000} respectively applied to $D_{validate}^{10,000}$.

Therefore, a quadratic-logistic discriminant can be utilized to have better fitting boundary, producing much lower error. There are shown below in Figure 4, producing the associated boundaries. Visually comparing these classification results to Figure 2, these results mirror generally the ideal MAP classification with full knowledge of the distribution. Furthermore, it can be seen in each case, that as the sample size increases, the error lowers and the decision boundaries more closely parallel the true distributions, which is intuitive as per the Law of Large Numbers, the classifier will converge upon the ideal classifier with the introduction of more training samples.

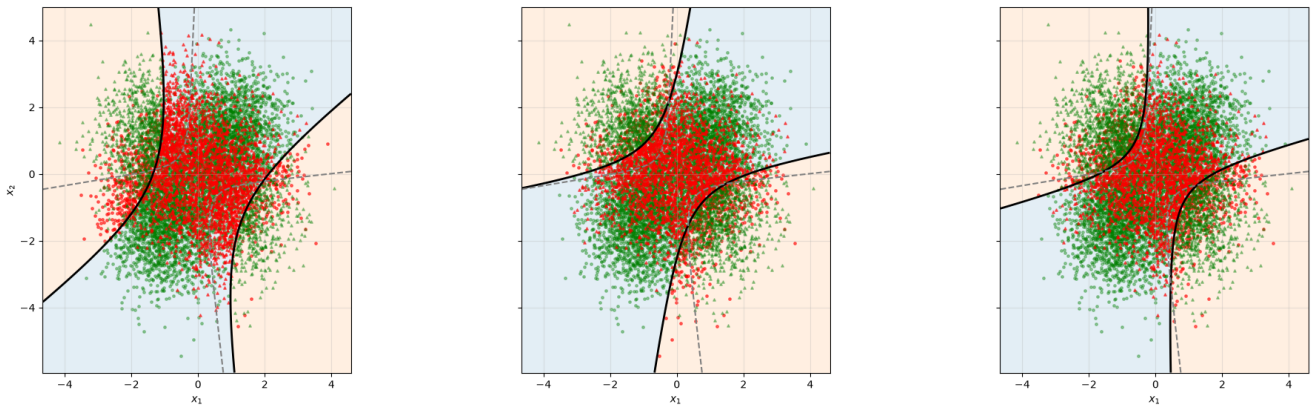


Figure 4: Logistic-Quadratic Classification Boundaries of D_{train}^{20}, D_{train}^{200}, and D_{train}^{2000} respectively applied to $D_{validate}^{10,000}$.

2 Question 2

For this question, a dataset was defined as $D = (\mathbf{x_1}, y_1), \ldots, (\mathbf{x}_N, y_N)$, where $y = c(\mathbf{x}, \mathbf{w}) + v$, where c(.,w) is a cubic polynomial in x with coefficients w and v is a random Gaussian random scalar with mean zero and σ^2-variance. Therefore, we must approximate the coefficients, W for these x,y pairs.

The coefficients, $\mathbf{w}$ defined below, are the target of optimization by the following techniques of Maximum Likelihood (ML) Parameter Estimation and Maximum A Posteriori (MAP) Parameter Estimation.

$$\mathbf{w} = [w_0, \quad w_1, \quad w_2, \quad w_3, \quad w_4, \quad w_5, \quad w_6, \quad w_7, \quad w_8, \quad w_9]$$

$$\mathbf{z}(\mathbf{x}) = [1, \quad x_1, \quad x_2, \quad x_1^2, \quad x_1x_2, \quad x_2^2, \quad x_1^3, \quad x_1^2x_2, \quad x_1x_2^2, \quad x_2^3]$$

2.1 Maximum Likelihood Parameter Estimation

To estimate the coefficients of the weight vector $\mathbf{w}$, the cubic polynomial can be written as $c(\mathbf{x}, \mathbf{w}) = \mathbf{w}^\top \mathbf{z}(\mathbf{x})$. Stacking the N training samples gives the design matrix $\mathbf{Z} \in R^{N\times 10}$, where each row is $\mathbf{z}(\mathbf{x}_n)^\top$, and the vector of observed targets $\mathbf{y}$. Since the noise is Gaussian, each target is distributed as $y_n \mid \mathbf{x}_n, \mathbf{w} \sim \mathcal{N}(\mathbf{w}^\top \mathbf{z}(\mathbf{x}_n), \sigma^2)$. Because the samples are iid, the likelihood is a product over the samples, and taking the negative logarithm yields Equation 4.

$$-\ln p(\mathcal{D} \mid \mathbf{w}) = \frac{N}{2}\ln(2\pi\sigma^2) + \frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{Z}\mathbf{w}\|^2 \tag{4}$$

Since we are optimizing for $\mathbf{w}$, the first term and the factor $1/(2\sigma^2)$ can be dropped, meaning ML estimation reduces to least squares. Taking the gradient, $-2\mathbf{Z}^\top\mathbf{y} + 2\mathbf{Z}^\top\mathbf{Z}\mathbf{w}$, and setting it to zero yields the closed-form solution in Equation 5. The noise variance is found the same way by minimizing Equation 4 with respect to σ^2, which is needed later for the MAP estimate.

$$\hat{\mathbf{w}}_{\mathrm{ML}} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{y}, \qquad \hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{N}\|\mathbf{y} - \mathbf{Z}\hat{\mathbf{w}}_{\mathrm{ML}}\|^2 \tag{5}$$

2.2 Maximum A Posteriori Parameter Estimation

For MAP estimation, a zero-mean Gaussian prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \gamma\mathbf{I})$ is placed on the weights, so the estimate maximizes $p(\mathcal{D} \mid \mathbf{w})\, p(\mathbf{w})$. Taking the negative logarithm and dropping the terms that do not contain $\mathbf{w}$ yields Equation 6.

$$\hat{\mathbf{w}}_{\mathrm{MAP}} = \arg\min_{\mathbf{w}} \ \frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{Z}\mathbf{w}\|^2 + \frac{1}{2\gamma}\|\mathbf{w}\|^2 \tag{6}$$

Unlike ML, the factor $1/(2\sigma^2)$ cannot be dropped here, since it only scales the data term and not the prior term. Taking the gradient, setting it to zero, and multiplying through by σ^2 yields the closed-form MAP estimate in Equation 7, where $\lambda = \frac{\sigma^2}{\gamma}$.

$$\hat{\mathbf{w}}_{\mathrm{MAP}} = \left(\mathbf{Z}^\top\mathbf{Z} + \lambda\mathbf{I}\right)^{-1}\mathbf{Z}^\top\mathbf{y} \tag{7}$$

Here, γ is the prior variance of each weight, and σ^2 is replaced by $\hat{\sigma}^2_{\mathrm{ML}}$ from Equation 5. This shows the relationship between the two estimators: the only change from ML to MAP is the added $\lambda\mathbf{I}$ term. As $\gamma \to \infty$, $\lambda \to 0$ and $\hat{\mathbf{w}}_{\mathrm{MAP}}$ converges to $\hat{\mathbf{w}}_{\mathrm{ML}}$, while as $\gamma \to 0$, $\hat{\mathbf{w}}_{\mathrm{MAP}}$ is forced toward $\mathbf{0}$. By conducting a sweep of γ from 10^{-15} to 10^{8}, the optimal γ was found at 1.70×10^{-6}, where the validation error is minimized.

2.3 Results

The results of these fits are described below in Table 2. It provides a descriptor of the error of the attempted classifiers as well as the coefficients of the corresponding equations that describe these cubic polynomial expressions. The experimental setup had a validation set $\mathcal{D}_{validate}$ of 1,000 samples and a training set, $\mathcal{D}_{train}$ of 100 samples.

Quantity	Value
Estimated noise variance $\hat{\sigma}^2_{\mathrm{ML}}$	2.8466
MAP best γ	1.70×10^{-6}
ML validation error	4.0660
MAP best validation error	3.8799

Table 2: ML and MAP results on $\mathcal{D}_{\mathrm{validate}}$ (1000 samples), trained on $\mathcal{D}_{\mathrm{train}}$ (100 samples).

$$\begin{aligned} c(\mathbf{x}, \hat{\mathbf{w}}_{\mathrm{ML}}) = {} & -0.2205 + 0.0948\, x_1 - 0.2596\, x_2 + 0.0005\, x_1^2 + 0.0222\, x_1x_2 \\ & + 0.1078\, x_2^2 - 0.0111\, x_1^3 - 0.0012\, x_1^2x_2 - 0.0001\, x_1x_2^2 + 0.0613\, x_2^3 \end{aligned} \tag{8}$$

$$c(\mathbf{x}, \hat{\mathbf{w}}_{\text{MAP}}) = -0.00001 - 0.00007\,x_1 - 0.00001\,x_2 - 0.00032\,x_1^2 + 0.00005\,x_1x_2 - 0.00982\,x_1^3 - 0.00017\,x_1^2x_2 - 0.00001\,x_1x_2^2 \quad (9)$$

Furthermore, the relationship between W_{ML} and W_{MAP} are described below in Figure 5. This figure is significant because it describes how the MAP validation error is tuned better reflecting the data. With tuning hyperparameters, there is always the risk of "fitting the noise", or overfitting, which is why each γ is validated on the test set after being fit to the training set. This in turn yields the "sweet spot" as seen in the left figure. The right figure demonstrates the convergent nature of MAP and ML Parameter Estimation, that being that as γ increases, MAP closer reflects pure ML estimates, which is intuitive from the only change to ML formula to MAP is the $\lambda\mathbf{I}$ term's addition.

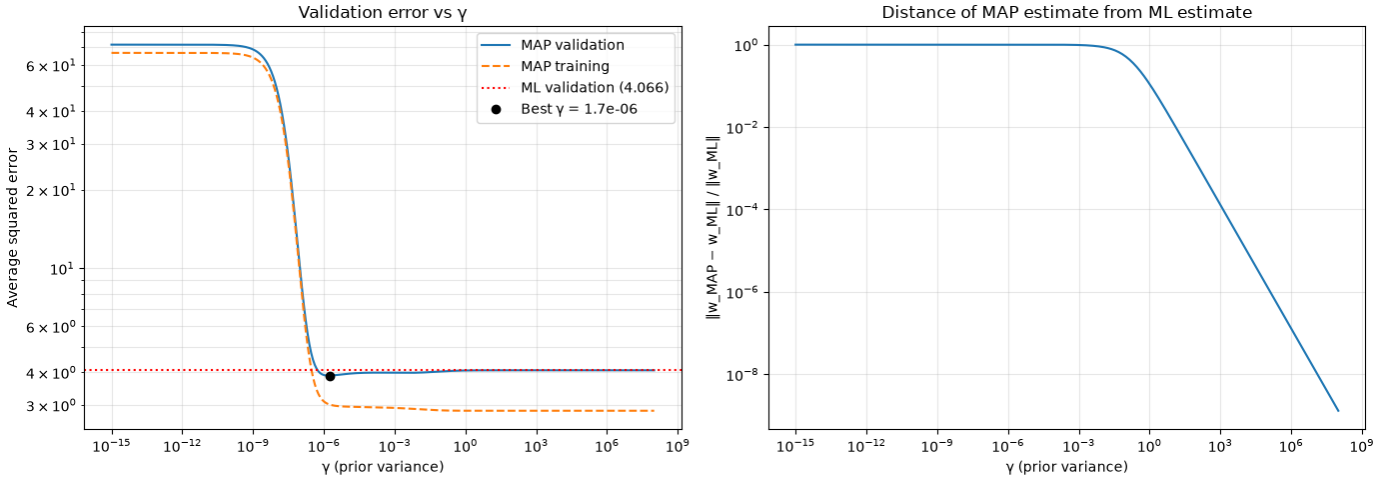


Figure 5: Validation error vs. γ, and MAP–ML convergence

3 Question 3

This optimization problem is prefaced on the following expressions which describe the position of the vehicle. $[x_T, y_T]^{\mathrm{T}}$ represents the true position of the car, while K landmarks at known positions measure range measurements estimating from those points. These measurements are given by $r_i = d_{Ti} + n_i$ where $d_{Ti} = ||[x_T, y_T]^T\text{-}[x_i, y_i]^T||$. The true distribution of the vehicle position is given below in Equation 10.

$$p\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = (2\pi\sigma_x\sigma_y)^{-1}\, e^{-\frac{1}{2}\begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix}^{-1}\begin{bmatrix} x \\ y \end{bmatrix}} \quad (10)$$

3.1 Optimization Equation Derivation

The MAP estimate is given by in Equation 11, which since each sample is i.i.d, can be represented as a product over the landmarks. Isolating the right side of the equation and applying Bayes, expands from Equation 11 to Equation 12. $P(x, y)$ and $P(r_i|x, y)$ can be substituted from Equation 10 and the general form of Gaussian with prior respectively, resulting in Equation 13, which applying a logarithm can be condensed to Equation 14 This results in the change of arg max to an arg min and the product to a summation.

$$\begin{bmatrix} x_{MAP} \\ y_{MAP} \end{bmatrix} = \arg\max_{x,y} P(x, y|r_1, \ldots, r_K) \quad (11)$$

$$= \arg\max_{x,y} \prod_{i=1}^{K} P(r_i|x, y)P(x, y) \quad (12)$$

$$= \arg\max_{x,y} \prod_{i=1}^{K} \left[(2\pi\sigma_i^2)^{-\frac{1}{2}}\, e^{-\frac{\left(r_i - d_i(x,y)\right)^2}{2\sigma_i^2}} \right] (2\pi\sigma_x\sigma_y)^{-1}\, e^{-\frac{1}{2}\begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix}^{-1}\begin{bmatrix} x \\ y \end{bmatrix}} \quad (13)$$

$$= \arg\min_{x,y} \sum_{i=1}^{K} \ln(\sqrt{2\pi\sigma_i^2} + \frac{(r_i - d_i(x_i, y_i))^2}{2\sigma_i^2} + \ln(2\pi\sigma_x\sigma_y) + \frac{1}{2}(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}) \tag{14}$$

Since we are optimizing for the position of (x, y), the term $\ln(2\pi\sigma_x\sigma_y)$, $\ln(\sqrt{2\pi\sigma_i^2})$, and $\frac{1}{2}$ can be dropped from the optimization problem. Now, subbing in for $d_i(x, y)$, the following optimized equation is derived.

$$\begin{bmatrix} x_{MAP} \\ y_{MAP} \end{bmatrix} = \arg\min_{x,y} \sum_{i=1}^{K} \frac{\left(r_i - \sqrt{(x - x_i)^2 + (y - y_i)^2}\right)^2}{\sigma_i^2} + \frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2} \tag{15}$$

3.2 Implementation

The true vehicle position is drawn uniformly inside the unit circle and using a fixed seed, this gave $[x_T, y_T]^\top = [-0.153, 0.488]^\top$. For each $K \in \{1, 2, 3, 4\}$, K landmarks are placed at angles $2\pi k/K$ on the unit circle, and each range is generated as the true distance plus Gaussian noise with $\sigma_i = 0.3$, redrawing any negative measurement. The prior uses $\sigma_x = \sigma_y = 0.25$, which expresses a fairly strong belief that the vehicle is near the origin.

The objective in Equation 15 is evaluated on a 401×401 grid spanning $[-2, 2]$ in each coordinate (spacing 0.01), for 160,801 candidate positions. The MAP estimate is the grid point with the lowest objective value. To allow visual comparison across K, all four objectives are computed first, and a single shared set of logarithmically spaced contour levels is used for every plot.

3.2.1 $K = 1$

Per Equation 15, the minimum value of x, y is found by subtracting the known range of the landmark to the target minus the euclidean distance of the target to the proposed location. If exactly on the mark, this would be equal to zero plus the prior term. This yields for $K = 1$ that an (x_{MAP}, y_{MAP}) that is on the "ring" of ranges surrounding the landmark with the lowest prior value.

3.2.2 $K = 2$

With 2 landmarks, the summation element of the MAP estimator means that the optimal point will be effectively where the two range rings of the landmarks intersecting, with the intersection with the lower of the two's prior term being favored. This leads in an increase in error, as this intersection effect causes for the landmarks to hit rather above or below the location of the vehicle.

3.2.3 $K = 3$

Three landmarks, operating with analogous characteristics to the two other cases, performs well as it effectively triangulates the vehicle, with each of the evenly spaced landmarks being able to closely approximate the true (x, y) with the MAP Estimator.

3.2.4 $K = 4$

The fourth landmark, when added, does not increase the performance of the MAP estimation. This is expected as the prior three landmarks were effective at minimizing the geometric limitations of the first two techniques by eliminating the picking a location on the ring as seen in $K = 1$ influenced by the prior or the over/undershooting of $K = 2$. $K = 4$ underperforms $K = 3$ due to the square formation of the sensors creating too much wiggle horizontally and vertically for the vehicle to be missed.

3.3 Results

Figure 6 below showcases the behavior of the four cases discussed in the prior section.

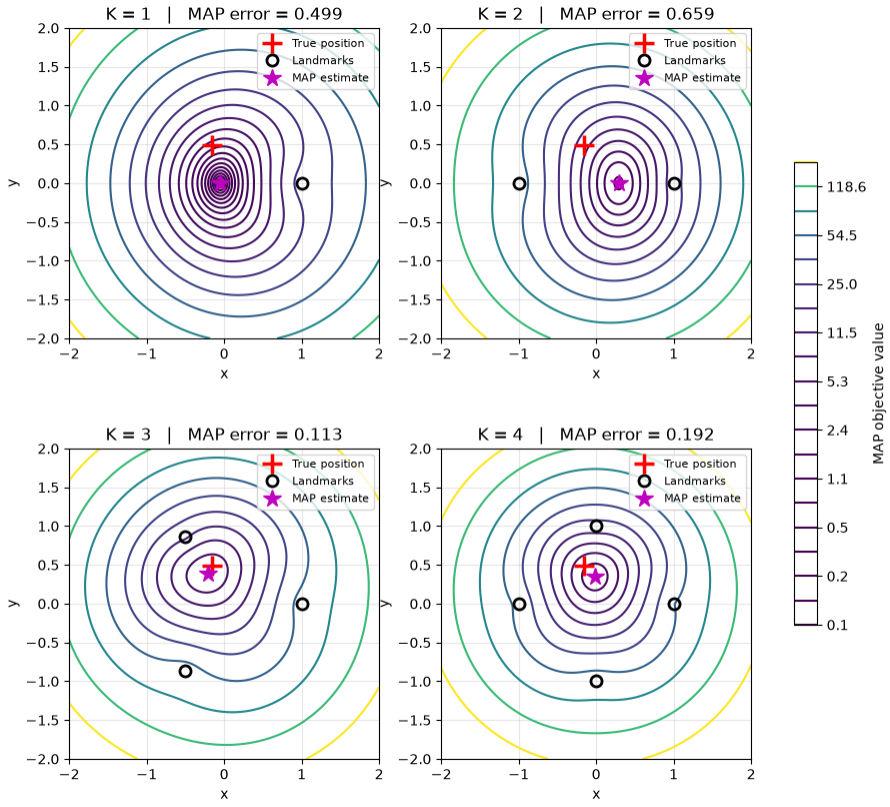


Figure 6: MAP Estimation of True Vehicle Position for $K \in \{1, 2, 3, 4\}$

Additionally, when experimenting with higher K values, the geometry theory tends to hold for why $K = 3$ has the highest effectiveness of the samples. It can be seen below in Figure 7 that performance of the MAP estimator does not get significantly better for the addition of more sensors past $K = 3$, validating that the triangulation effect on being an effective means of optimizing the MAP estimation.

It is important to note that over the course of this experiment, the results become more certain as the number of landmarks increases. Even though $K = 4$ has a slightly larger error than $K = 3$, the certainty still rises and the error trends downward overall. This is further supported by the tightening of the contours around the MAP estimate, which indicates a greater degree of certainty at higher values of K.

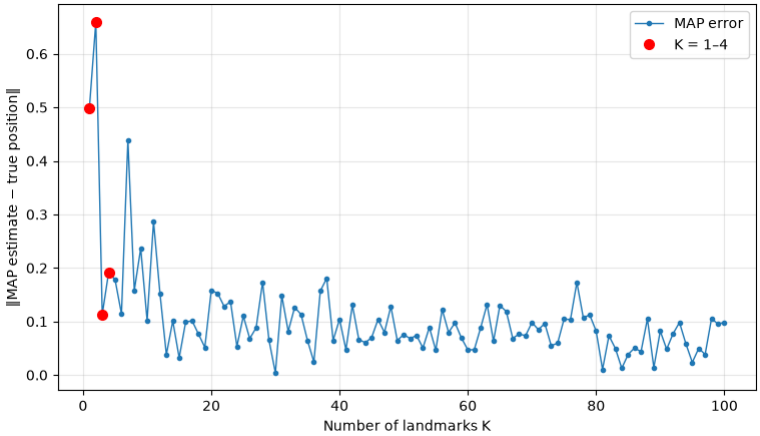


Figure 7: MAP Estimation Error for $K \in \{1, ..., 100\}$

4 Question 4

Introduced is the concept of adding an additional option to classification problems of a "rejection", where if a data point look dissimilar from any class in a dataset, it is rejected versus potentially misclassifying that data point. This is done through establishing the following loss functions.

$$\lambda(\alpha_i \mid \omega_j) = \begin{cases} 0 & i = j \qquad i, j = 1, \ldots, c \\ \lambda_r & i = c + 1 \\ \lambda_s & \text{otherwise} \end{cases}$$

Expected Risk Minimization seeks to create a decision rule which decides a class based upon the option which yields the lowest this. This general formula for risk is defined in Equation 16.

$$R(\alpha_i|x) = \sum_{j=1}^{c} \lambda_{ij} P(\omega_j|x) \tag{16}$$

Therefore, for ERM, risk is to be minimized so the appropriate class is chosen from the qualifying options. The is seen in the description of ERM, Equation 17.

$$\arg\min_i R(\alpha_i|x) \text{ for all i} \in \{1, ..., c+1\} \tag{17}$$

Therefore, when assessing the risk associated a rejection versus a substitution. We can derive the cost of substitution through the following Equations 18 and 19, which describes the cost of all values of the give actions associated with each class. The use of a summation allows certain terms to be removed from the summation. The cost value, λ, is defined for each of the decisions, those being correct classification, incorrect classification, λ_s, and the rejection case, λ_r. The risk function, being a sum, can be expanded to separate the correct classification from the whole of the incorrect classifications.

$$R(\alpha_i|x) = \lambda_s \sum_{j \neq i} P(\omega_j|x) = \lambda_s \big(1 - P(\omega_i|x)\big), \quad i \leq c \tag{18}$$

$$R(\alpha_{c+1}|x) = \lambda_r \sum_{j=1}^{c} P(\omega_j|x) = \lambda_r \tag{19}$$

Therefore, rejecting is preferred when its risk is lower than that of the best classification, the following inequality can be created following the minimization formula of ERM, where ω_i represents the class with the highest posteirior.

$$\lambda_r < \lambda_s(1 - P(\omega_i|x)) \tag{20}$$

This can be simplified further to derive the association between the two decision costs.

$$\lambda_r < \lambda_s - \lambda_s P(\omega_i|x) \tag{21}$$

$$\lambda_r - \lambda_s < -\lambda_s P(\omega_i|x) \tag{22}$$

$$1 - \frac{\lambda_r}{\lambda_s} > P(\omega_i|x) \tag{23}$$

Therefore, decide ω_i if $P(\omega_i|x) \geq 1 - \frac{\lambda_r}{\lambda_s}$; otherwise, reject.

4.1 Implications of Cost Functions

In this classification setup, the loss of each classification follows that $\lambda_r < \lambda_s$. This is important intuitively, as we should assume that is costs the least to classify correctly, then it should cost more to reject a sample, but it should be even more expensive to misclassify a sample. This means the cost of each λ should be very deliberate to maintain this order of costs. The improper cost assignment will create the following behaviors.

4.1.1 $\lambda_r = 0$

If there is no cost of rejection, λ_r, this would create the above expression defined in Equation 23 defined as the following.

$$1 > P(\omega_{i^*}|x) \tag{24}$$

Based upon Equation 24, this means that for all potential PDFs, the decision rule will be decided by if 1 is greater than that PDF. Since this is always true, it would follow that for every evaluation of this inequality, it would be found to be true, meaning that every classification would result in a rejection. This makes intuitive sense, since the lower the cost of rejection, λ_r, the more attractive it is to reject a sample.

4.1.2 $\lambda_r > \lambda_s$

Drawing again from Equation 23, setting $\lambda_r > \lambda_s$ would result in the following inequality as the decision rule.

$$-\alpha > P(w_i|x) \text{ where } \alpha \text{ is a non-negative number} \tag{25}$$

Based upon Equation 25, the decision rule becomes that for all PDFs, it would decide false. The comparison between λ_r and λ_s would always be some negative numver, which would never be greated than acceptance. Therefore, it would never theoretically reject a sample, making the term useless and unnecessary bloat upon the classification model, making the dropping of λ_r of no consequence.

5 Question 5

This question poses a Dirichlet distribution of categorical data which the dataset $\mathcal{D} = \{\mathbf{z}_1, \ldots, \mathbf{z}_N\}$ with i.i.d samples $\mathbf{z}_n \sim \text{Cat}(\Theta)$ for $n \in \{1, \ldots, N\}$. With such a distribution, parameter estimation depends upon the categorical data.

5.1 Maximum Likelihood Parameter Estimation

For ML estimation, Equation 26 is used as the basis for approximating the value Θ. Since the values are iid, the pdf can be treated as joint products to which the logarithm can be applied.

$$\hat{\Theta}_{\text{ML}} = \arg\max_{\Theta}\ p(\mathcal{D} \mid \Theta) = \arg\max_{\Theta} \prod_{n=1}^{N} p(\mathbf{z}_n \mid \Theta) = \arg\max_{\Theta} \sum_{n=1}^{N} \ln p(\mathbf{z}_n \mid \Theta) \tag{26}$$

From Equation 26, we can substitute in the equation for the probability mass function of a categorical distribution and apply a logarithm to the function, yielding the result seen in Equation 27.

$$p(\mathbf{z}_n \mid \Theta) = \prod_{k=1}^{K} \theta_k^{z_{nk}} \quad \Longrightarrow \quad \hat{\Theta}_{\text{ML}} = \arg\max_{\Theta} \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} \ln \theta_k \tag{27}$$

From Equation 27, the order of the summation can be exchanged and the count can be defined as $N_k = \sum_{n=1}^{N} z_{nk}$, so that $\sum_k N_k = N$. The log-likelihood becomes Equation 28.

$$\sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} \ln \theta_k = \sum_{k=1}^{K} N_k \ln \theta_k \tag{28}$$

Since the parameters satisfy $\sum_k \theta_k = 1$, a Lagrange multiplier λ can be introduced, creating Equation 29. Taking the derivative of this equation with respect to θ_k, yielding Equation 30. Reapplying the constraint $\sum_k N_k/\lambda = N/\lambda = 1$, so $\lambda = N$ yeilds the final solution, described in Equation 31.

$$\mathcal{L}(\Theta, \lambda) = \sum_{k=1}^{K} N_k \ln \theta_k + \lambda \left(1 - \sum_{k=1}^{K} \theta_k\right) \tag{29}$$

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = \frac{N_k}{\theta_k} - \lambda = 0 \quad \Longrightarrow \quad \theta_k = \frac{N_k}{\lambda} \tag{30}$$

$$\hat{\theta}_{k,\text{ML}} = \frac{N_k}{N} \tag{31}$$

5.2 Maximum A Posteriori Parameter Estimation

For MAP estimation, the parameters are chosen to maximize the posterior $p(\Theta \mid \mathcal{D})$ rather than the likelihood. Applying Bayes' rule and dropping $p(\mathcal{D})$, which does not depend on Θ, the MAP estimate becomes the product of the likelihood and the prior. Since the samples are iid, the likelihood factors into a product over samples, and applying the logarithm converts the products into sums, as shown in Equation 32.

$$\hat{\Theta}_{\text{MAP}} = \arg\max_{\Theta}\ p(\Theta \mid \mathcal{D}) = \arg\max_{\Theta}\ p(\mathcal{D} \mid \Theta)\, p(\Theta \mid \boldsymbol{\alpha}) = \arg\max_{\Theta}\ \sum_{n=1}^{N} \ln p(\mathbf{z}_n \mid \Theta) + \ln p(\Theta \mid \boldsymbol{\alpha}) \tag{32}$$

Taking the logarithm of the Dirichlet prior yields Equation 33. The normalization constant $B(\boldsymbol{\alpha})$ depends only on the hyperparameters and not on Θ, so the term $-\ln B(\boldsymbol{\alpha})$ does not affect the maximizer and can be dropped.

$$\ln p(\Theta \mid \boldsymbol{\alpha}) = -\ln B(\boldsymbol{\alpha}) + \sum_{k=1}^{K} (\alpha_k - 1) \ln \theta_k \tag{33}$$

The log-likelihood term was already simplified in Equation 28 to $\sum_k N_k \ln \theta_k$. Adding the remaining prior term and combining the sums over k gives the log-posterior in Equation 34.

$$\hat{\Theta}_{\text{MAP}} = \arg\max_{\Theta}\ \sum_{k=1}^{K} (N_k + \alpha_k - 1) \ln \theta_k \tag{34}$$

Equation 34 has the same form as the ML log-likelihood, with N_k replaced by $N_k + \alpha_k - 1$. Enforcing the constraint $\sum_k \theta_k = 1$ with a Lagrange multiplier λ creates Equation 35. Taking the derivative with respect to θ_k and setting it to zero yields Equation 36.

$$\mathcal{L}(\Theta, \lambda) = \sum_{k=1}^{K} (N_k + \alpha_k - 1) \ln \theta_k + \lambda \left(1 - \sum_{k=1}^{K} \theta_k\right) \tag{35}$$

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = \frac{N_k + \alpha_k - 1}{\theta_k} - \lambda = 0 \quad \Longrightarrow \quad \theta_k = \frac{N_k + \alpha_k - 1}{\lambda} \tag{36}$$

Reapplying the constraint gives $\sum_k (N_k + \alpha_k - 1)/\lambda = \left(N + \sum_k \alpha_k - K\right)/\lambda = 1$, so $\lambda = N + \sum_k \alpha_k - K$. Substituting back yields the final solution, described in Equation 37.

$$\hat{\theta}_{k,\text{MAP}} = \frac{N_k + \alpha_k - 1}{N + \sum_{j=1}^{K} \alpha_j - K} \tag{37}$$

This estimate is the mode of the posterior and is valid when $N_k + \alpha_k > 1$ for all k. The quantities $\alpha_k - 1$ act as pseudo-counts added to the observed counts. For a uniform prior ($\alpha_k = 1$ for all k), the MAP estimate reduces to the ML estimate N_k/N, and as $N \to \infty$ the observed counts dominate the pseudo-counts, so $\hat{\theta}_{k,\text{MAP}}$ converges to $\hat{\theta}_{k,\text{ML}}$.

Acknowledgment of AI Assistance

Claude (Anthropic), an AI assistant, was used extensively throughout this assignment. Claude explained the underlying theory, including expected risk minimization and the minimum-P(error) likelihood-ratio test, ROC curves and discriminant scores, maximum likelihood training of logistic-linear and logistic-quadratic class-posterior models, gradient descent and step-size selection, ML and MAP parameter estimation for linear-in-parameters regression (including the connection to ridge regression and the role of γ and $\lambda = \sigma^2/\gamma$), the reject option in Bayesian decision theory, range-based localization and trilateration, and the categorical and Dirichlet distributions. It also pointed me to relevant sections of the course notes and lecture slides.

Claude reviewed and debugged my code, identifying issues including swapped TPR/FPR assignments and an inverted threshold γ in my ROC code, the use of a non-log-likelihood-ratio score, an incompatible `float()` conversion and deprecated `Axes3D` call in the provided data generation script, transposition of the generated data, misuse of `scipy.optimize.minimize` (passing a function value instead of the function), a MAP routine that called the ML optimizer, validation features passed as training targets, and an overwritten results dictionary in the landmark-count experiment. Claude also wrote substantial portions of the code, including the gradient descent training loop for the logistic models, the decision-boundary visualizations, the closed-form ML/MAP estimators and γ sweep for Question 2, the simulation, MAP objective, and contour plotting code for Question 3, and the error-versus-K plot. I integrated, ran, and adapted this code, and all reported numerical results were produced by running it.

In addition, Claude provided LaTeX formatting for equations, tables, and figures; corrected errors in my derivations for Questions 1 through 5; drafted or rewrote portions of the report text, including the Question 1 parameter estimation section (Section 1.2), the Question 2 derivations (Sections 2.1 and 2.2), the Question 3 MAP objective derivation (Section 3.1), implementation description (Section 3.2), and certainty discussion, corrections to the Question 4 reject-option derivation, and the Question 5 ML and MAP derivations (Sections 5.1 and 5.2); and reviewed drafts against the assignment requirements. The transcript for this session can be found here: `https://claude.ai/share/77cac50c-470f-416e-afcd-4aba2d1e7813`

All code used in this project, as well as figures, can be found at: `https://github.com/RonanEngel/EC5644_Assignments`